# Quantizable Models for M-theory

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(Based on work with Warren Siegel [1, 2, 3, 4].)

## S-theory

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- Green & Schwarz (1984) [5]: 10D supergravities "$S(X)$" on spacetime $X$ originate from quantization of fundamental strings "$\mathbf{S}(X)$" with target $X$.
- Moduli space of Type-II supergravity $S(X)$ on $X \to X \times T^n$:

$$\frac{O(n,n)}{O(n) \times O(n)}$$

  due to the symmetry between momentum modes (one copy of $n$) and winding modes (the other copy of $n$).
- Narain (1986) [6]: $O(n,n)$ symmetry of $S(X)$ broken to $O(n,n;\mathbf{Z})$ "T-duality" of the superstring $\mathbf{S}(X)$.

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$$(\partial_M) := \begin{pmatrix} \partial_\mu \\ \tilde{\partial}^\mu \end{pmatrix} \quad \eta := \begin{pmatrix} 0 & \delta^\nu_\mu \\ \delta^\mu_\nu & 0 \end{pmatrix} \quad \partial^M \partial_M = 0 \quad \Rightarrow \quad \text{e.g.} \quad \tilde{\partial}^m = 0.$$

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Extension to a string theory $\mathbf{T}(X)$ with manifest T-duality symmetry. New Lie derivative (D-bracket) and commutator (C-bracket) described by current algebra of $O(D, D)$-covariant "generalized" momenta $\Pi_M = P_M + \eta_{MN} X'^N$ with quantum bracket

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$$[\Pi_M(\sigma_1), \Pi_N(\sigma_2)] = 2i\eta_{MN}\delta'(\sigma_1 - \sigma_2).$$

Section condition arises dynamically as a new $O(D,D)$-invariant Virasoro constraint

$$L = \tfrac{1}{2}\eta^{MN}\Pi_M\Pi_N \quad \Rightarrow \quad \partial^M\partial_M = 0.$$

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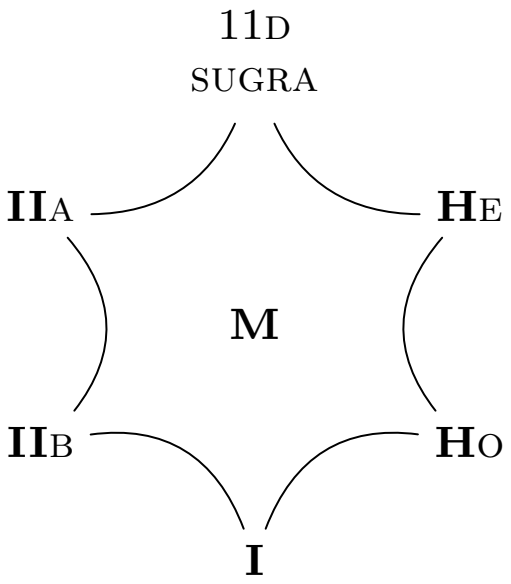
- Witten (1995) [14]: S-duality of the type-IIA **S**tring theory has 11-dimensional supergravity as its massless truncation.



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- Cremmer & Julia (1979) [15] and Hull & Townsend (1994) [16]: Analogously to the moduli space of compactifications of the type-II string theories on $n$-tori, hidden *exceptional* "U-duality" symmetry $E_{n(n)}$ of (gauged) 11D supergravity manifested by compactification on $T^n$.

| $n$ | $E_{n(n)}$ | $H_n$ | # vectors | # scalars |
|---|---|---|---|---|
| 3 | $SL(2;\mathbf{R}) \times SL(3;\mathbf{R})$ | $SO(2) \times SO(3)$ | 6 | 7 |
| 4 | $SL(5;\mathbf{R})$ | $SO(5)$ | 10 | 14 |
| 5 | $Spin(5,5)$ | $SO(5) \times SO(5)$ | 16 | 25 |
| 6 | $E_{6(6)}$ | $Sp(4)$ | 27 | 42 |
| 7 | $E_{7(7)}$ | $SU(8)$ | 28 | 70 |
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- Duff (1985) [17]: "Either the $E_8 \times SO(16)$ structure appears only in $d = 3$ as an <u>internal</u> symmetry, or else it is really present at every $d \leq 11$ as a <u>spacetime</u> symmetry which transforms states of different spin into each other, in particular transforming [the frame field] into [the 3-form] in $d = 11$. We emphasize that we are advocating the latter."

# F-theory?

Vafa (1996) [18]: $SL(2;\mathbf{Z})$ S-duality of the type-IIB **S**tring may be realized geometrically in "12D" theory $F$

$$\begin{tikzcd}
 & F(\widehat{X}) \arrow[dl, dotted] \arrow[dd] \\
M(X \times S^1) \arrow[dr] & \\
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Defining properties of $F$:

1. Geometrically realized (S(trong/weak) part of) U-duality
2. Not really 12D: Type-IIB **S**tring with axio-dilaton profile determined by elliptically-fibered Calabi-Yau 4-fold $T^2 \to \widehat{X} \to X$
3. Related by compactification to M-theory

# Definition of $F(X)$

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# Definition of $F(X)$

We propose to generalize the definition of F-theory by using both doubled and exceptional geometry [1]:
For any $X$, with supergravity $S(X)$, M-gravity $M(X)$, and generalized & doubled gravity $T(X)$, define $F(X)$ to make the following diagram commute:[1]

$$\begin{tikzcd}
 & \begin{array}{c} F(X) \\ E_{n(n)}/H_{n(n)} \end{array} \arrow[dl] \arrow[dr] & \\
\begin{array}{c} M(X) \\ \frac{GL(D+1,\mathbf{R})}{O(D,1)} \end{array} \arrow[dr, "\text{compactification}"'] & & \begin{array}{c} T(X) \\ \frac{O(D,D)}{O(D)\times O(D)} \end{array} \arrow[dl, "\text{sectioning}"] \\
 & \begin{array}{c} S(X) \\ \frac{GL(D,\mathbf{R})}{O(D-1,1)} \end{array} &
\end{tikzcd}$$

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# Extention to fundamental theories

Goal: For each rank $n = D + 1$ find brane theory (generalizing the string), the quantization of which gives the theory $\mathbf{F}(X)$ that completes the diagram

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 & \mathbf{F}(X) \arrow[dl, "\text{section}"'] \arrow[dr, "\text{reduction}"] & \\
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It is *a priori* unclear whether this problem (existence of $\mathbf{F}$) has a solution.
In fact, we have the following

## Result ([2, 3, 4])

*Let $D = \dim(X) \leq 6$. Then $\mathbf{F}(X)$ exists in the form of a current algebra with exceptional global symmetry $G = E_{n(n)}$ of rank $n = D + 1$. This current algebra arises from the quantization of a free theory with Hamiltonian symmetry $H_{n(n)}$ the (split form of the) maximal compact subgroup of $E_{n(n)}$. When a Lagrangian formulation is known ($n \leq 5$), the symmetry is enhanced to $L_{n(n)}$ and the theory is that of a free self-dual form on a brane of split signature.*

| $n$ | Lagrangian $L_{n(n)}$ | Hamiltonian $H_{n(n)}$ | Current Algebra $E_{n(n)}$ |
|---|---|---|---|
| 2 | $SL(2;\mathbf{R})$ | $SO(1,1)$ | $GL(2;\mathbf{R})$ |
| 3 | $SL(2;\mathbf{R})^2 \times SO(2)$ | $SL(2;\mathbf{R}) \times SO(2)$ | $SL(3;\mathbf{R}) \times SL(2;\mathbf{R})$ |
| 4 | $SL(4;\mathbf{R}) \simeq Spin(3,3)$ | $Sp(4;\mathbf{R}) \simeq Spin(3,2)$ | $SL(5;\mathbf{R})$ |
| 5 | $SL(4;\mathbf{C}) \simeq Spin(6;\mathbf{C})$ | $Sp(4;\mathbf{C})$ | $Spin(5,5)$ |
| 6 | ¿ $SU^*(8)$ ? | $USp(4,4)$ | $E_{6(6)}$ |
| 7 | ¿ $U^*(8)$ ? | $SL(4;\mathbf{H})$ | $E_{7(7)}$ |

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- F-gravity $F(X)$ has exceptional U-duality symmetry of rank 4: $G = E_{4(4)} \simeq SL(5;\mathbf{R})$.
- Coordinates $X^{mn}$ in the **10** (signature 4+6). But solution of sectioning

$$\varepsilon^{mnpqr}\partial_{mn}\partial_{pq} = 0 \quad \Rightarrow \quad (\partial_{mn}) = (\partial_{-1\mu}, \partial_{\mu\nu}) \to (\partial_\mu, 0)$$

  reduces this to $(3+1)$-dimensional $M(X)$ breaking $SL(5;\mathbf{R}) \to GL(4;\mathbf{R})$ spontaneously [20].

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- So we postulate for $\mathbf{F}(X)$ at each fixed $\tau$ a wordvolume with 5 coordinates $\sigma^m$ (in addition to the evolution coordinate $\tau$) and $G = SL(5;\mathbf{R})$-covariant current

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- Implies $\varepsilon^{mnpqr}(\partial_{mn} f)(\partial_{pq} g) = 0$ for all functions $f$ and $g$ on $F(X)$.

# Problem ... and solution

But now algebra does not close:

$$[\mathcal{S}^m, \mathcal{S}^n] = \mathcal{S}^{(m}\,\partial^{n)}\delta + \partial^{[m}\mathcal{S}^{n]}\,\delta + \varepsilon^{mnpqr}\Pi_{pq}\mathcal{U}_r\,\delta \quad \text{with} \quad \mathcal{U}_r := \partial^s\Pi_{rs}.$$

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... but this is a Gauß law constraint! Analogously to the first-quantization of Maxwell theory [21]: Quantize a 6D gauge 2-form $X_{MN}$!

| | gauge field | Lagrange multiplier | dynamical components | conjugate momentum | Gauß's law |
|---|---|---|---|---|---|
| Maxwell | $A_\mu$ | $A^\tau$ | $A^i$ | $E^i$ | $\partial^i E_i$ |
| $\mathbf{F}(X)$ | $X_{MN}$ | $X^{\tau m}$ | $X^{mn}$ | $P_{mn}$ | $\partial^n P_{mn}$ |

# Chiral forms

Siegel (1983) [22]: By analogy with string action for chiral scalar fields, action for gauge 2-form with self-dual fieldstrength $F_{MNP}$:

$$S_{iegel} = \int \left( \frac{1}{12} F^{MNP} F_{MNP} + \frac{1}{4} \lambda^{MN} F_M^{(-)PQ} F_{NPQ}^{(-)} \right) d^6\sigma.$$

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implies the fieldstrength of $X$ is self-dual. The gauge transformations

$$\delta X_{MN} = \alpha^P F^{(-)}_{MNP} + \tfrac{1}{12} \alpha^P \lambda_{[M}{}^Q F(-)_{NPQ]} + \mathcal{O}(\lambda^2)$$
$$\delta \lambda^{MN} = \partial^{(M} \alpha^{N)} + \alpha^P \partial_P \lambda^{MN} + \tfrac{1}{2} \lambda_P{}^{(M} \left( \partial^{N)} \alpha^P - \partial^P \alpha^{N)} \right) - \text{trace} + \mathcal{O}(\lambda^2)$$

imply $\lambda$ is pure gauge.

## Canonical analysis

In $\lambda \to 0$ gauge the Hamiltonian action becomes

$$S = -\int \tfrac{1}{2}\dot{X}^{mn} P_{mn}\, d^5\sigma d\tau + \int H\, d\tau$$

$$H = \int \left(\tfrac{1}{4}P^{mn}P_{mn} + \tfrac{1}{12}F^{mnp}F_{mnp} + X^{\tau m}\mathcal{U}_m\right)\, d^5\sigma.$$

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Free quantum brackets $[P_{mn}(\sigma_1), X^{pq}(\sigma_2)] = -i\delta^{[p}_{[m}\delta^{q]}_{n]}\delta(\sigma_1 - \sigma_2)$ imply

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$$\mathcal{T}_{mn} = \eta_{mn}\mathcal{T} - \tfrac{1}{2}\eta^{pq}\Pi_{mp}\Pi_{nq} \qquad \boxed{\mathcal{S}^r = \tfrac{1}{16}\epsilon^{mnpqr}\Pi_{mn}\Pi_{pq}} \qquad \mathcal{T} = \tfrac{1}{8}\eta^{mp}\eta^{nq}\Pi_{mn}\Pi_{pq}$$

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$\Rightarrow$ Closed $E_{4(4)} = SL(5;\mathbf{R})$-covariant sub-algebra generated by $\mathcal{S}$ and $\mathcal{U}$!

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Spacetime gauge transformations generated by vector field currents

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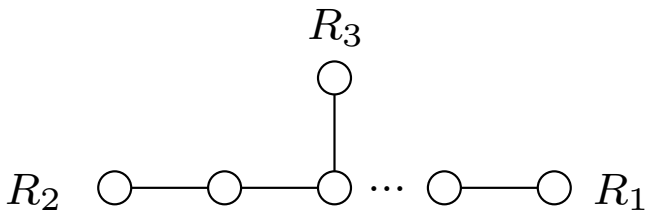

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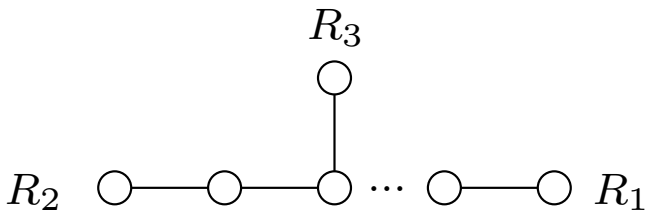


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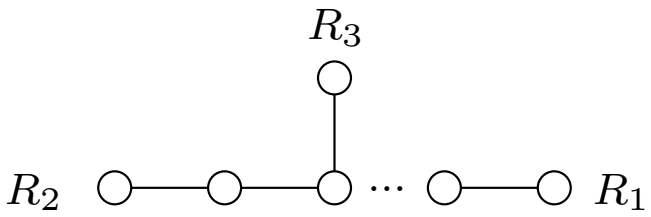


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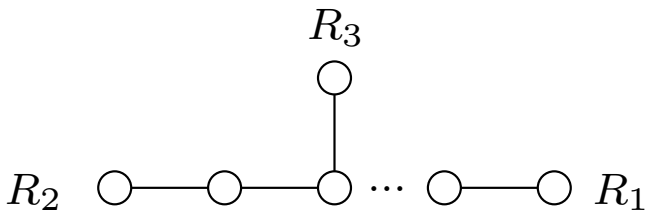


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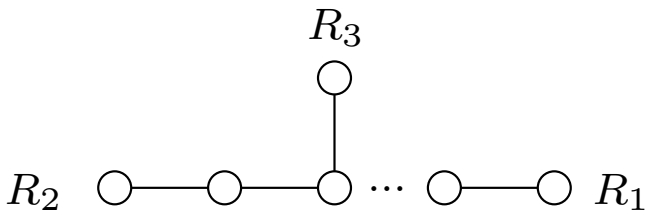


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Ultimate goal is 10-dimensional Lorentz invariance for $X$ corresponding to $E_{11}$ U-duality conjectured by West (2001) [24].

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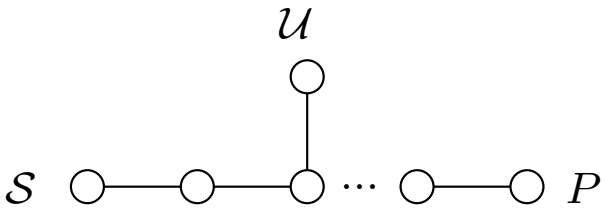

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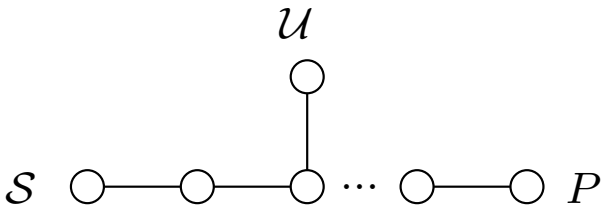


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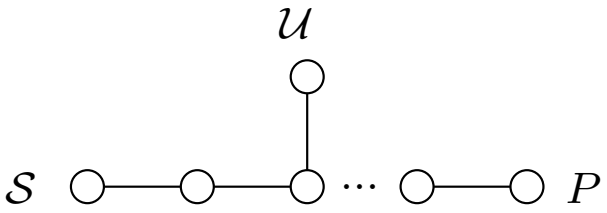


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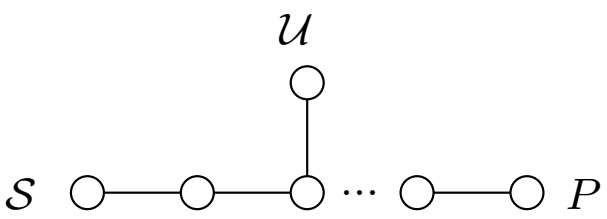


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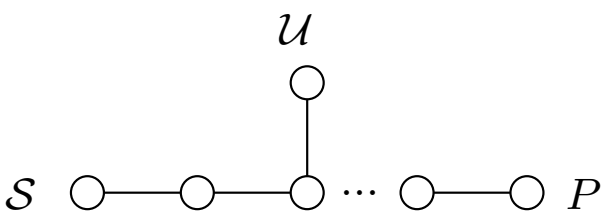


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